

1) Cipher text: HLDTQIHJDWXQCMAG

Key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ find plain text

determinant

$$|K| = (3 \times 7) - (2 \times 5) = 21 - 10 = 11$$

inverse of determinant

$$11^{-1} \text{ mod } 26$$

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3(1)$$

$$1 = 4 - (11 - 4(2))$$

$$1 = 4 - 11 + 4(2)$$

$$1 = 3(4) - 11$$

$$1 = 3(26 - 11(2)) - 11$$

$$1 = 3(26) - 11(6) - 11$$

$$1 = 3(26) - 11(7)$$

$$\Rightarrow 26 - 7$$

$$\Rightarrow 19$$

HL|DT|QI|HJ|DX|WQ|CM|AG
7 11 3 19

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(7) + (14 \times 11) \\ (9 \times 7) + (5 \times 11) \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3(3) + (14 \times 19) \\ 9(3) + (5 \times 19) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 275 \\ 122 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = \begin{bmatrix} P \\ S \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 16 + (14 \times 8) \\ 9 \times 16 + (5 \times 8) \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 160 \\ 184 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} F \\ C \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 4 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 7 + 14 \times 4 \\ 9 \times 7 + 5 \times 4 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 147 \\ 108 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{bmatrix} R \\ F \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 3 + 14 \times 23 \\ 9 \times 3 + 5 \times 23 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 331 \\ 142 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{bmatrix} T \\ M \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 22 + 14 \times 16 \\ 9 \times 22 + 5 \times 16 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 290 \\ 271 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{bmatrix} F \\ S \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 2 + 14 \times 12 \\ 9 \times 2 + 5 \times 12 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 174 \\ 78 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 18 \\ 6 \end{bmatrix} = \begin{bmatrix} S \\ A \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 \times 0 + 14 \times 6 \\ 9 \times 0 + 5 \times 6 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 84 \\ 30 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} G \\ F \end{bmatrix}$$

PT $\Rightarrow$ TOP SECRET MESSAGE

2) Consider the Hill cipher Key Matrix

$$K = \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} \quad \text{Cipher Text: RXAODY} \quad \text{Find plain text}$$

$$\Delta \begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} = (24 \times 14) - (5 \times 19) = 241$$

$$241 \bmod 26 = 7$$

$$7^{-1} \Rightarrow 26 = 7(3) + 5 \Rightarrow 5 = 26 - 7(3)$$

$$7 = 5(1) + 2 \Rightarrow 2 = 7 - 5(1)$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 2(7 - 5(1))$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = (-11)7 + 26(3)$$

$$\Rightarrow 26 - 11 = 15$$

$$15 \begin{bmatrix} 14 & 19 \\ -5 & 24 \end{bmatrix} = \begin{bmatrix} 210 & -285 \\ -75 & 370 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$i) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 2 \times 17 + 1 \times 23 \\ 3 \times 17 + 22 \times 23 \end{bmatrix} = \begin{bmatrix} 547 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} F \\ L \end{bmatrix}$$

$$ii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 2 \times 0 + 1 \times 14 \\ 3 \times 0 + 22 \times 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{bmatrix} O \\ W \end{bmatrix}$$

$$iii) \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 2(3) + 1(24) \\ 3(3) + 22(24) \end{bmatrix} \bmod 26 = \begin{bmatrix} 30 \\ 531 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{bmatrix} E \\ R \end{bmatrix}$$

$$\text{PT} = \underline{\text{FLOWER}}$$

3) Relation b/w substitution, confusion, transposition and Diffusion.

i) Substitution & confusion:

: Substitution replaces one character with another character.

Eg: HELLO $\rightarrow$ KHOOR

The relationship is:

Substitution creates confusion by hiding the relationship between original plain text characters & ciphertext characters.

Substitution $\rightarrow$ Confusion:

: Makes it difficult to understand how the key and plaintext are related to the ~~plain~~ cipher text.

ii) Transposition → Diffusion:

Transposition changes the position of characters w/o changing the characters themselves.

eg: HELLO → LHOEL

The relationship:

Transposition creates diffusion by spreading the influence of plaintext characters across different positions in ciphertext.

Transposition → Diffusion:

Diffusion spreads info of plaintext throughout the ciphertext making pattern harder to recognize.

Cipher	Does it change the character / symbol value?	Does it change character position	Confusion / Diffusion	Justification
Shift cipher	yes	No	Confusion	Replaces each character with a different character based on an addition key, obscuring the PT without moving character locations.
Multiplicative cipher	yes	No	Confusion	Replaces each character with another character using a mathematical multiplication modulo 26, creating substitutions without moving positions.
Affine cipher	yes	No	Confusion	Combines both shift & multiplicative principles to substitute characters, entirely focusing on altering the symbols value to hide relationship.
Transposition cipher	No	yes	Diffusion	Keeps all original characters but scatters them to different positions, spreading the linguistic pattern across the cipher text.

Transposition

Encryption

Encrypt the message = "The arrow is pointing north direction" using
= Simple
with key = 4

- Keyword

with keyword = ATLAS

- double

Key1 = BREAK

Key2 = Log

i) Simple (key = 4)

1	2	3	4
T	H	E	A
R	R	O	W
I	S	P	O
I	N	T	I
N	G	N	O
R	T	H	O
I	R	E	C
T	I	O	N

Cipher Text:

TRIINRIT HRSNGTRI EOPINHEO
AWQIODCN

ii) Keyword (key = ATLAS)

①	⑤	③	②	④
A	T	L	A	S
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	x	x	x

CIPHER TEXT:

TRPIRRO AINNDT EWIGHCK RSTOIX
HOONTEN

iii) Double Transposition (key1 = BREAK, key2 = LOG)

②	⑤	③	①	④
B	R	E	A	K
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	x	x	x

CIPHER TEXT:

AINNDTX TRPIRRO EWIGHCK RSTOIX
HOONTEN

LOG

AIN
NOT
XTR
PIR
ROE
WIG
HCX
EST
OSI
XHO
ONT
ENX

CIPHER TEXT:

ANXPRWHROXOE IOTIGICSHNON
NTRREGXTIOTK

Decryption:

TIDIE MRNME REOGO SANOW RSLTP EEEEO SSSNO AHTON
BIENP GODAE PEIXD ELNSX

→ simple with key = 10

$$R = \frac{60}{10} = 6$$

R1	T	R	O	O	P	S	H	E	A	D
R2	I	N	G	W	E	S	T	N	E	E
R3	D	M	O	R	E	S	U	P	P	L
R4	I	E	S	S	E	N	D	G	E	N
R5	E	R	A	L	D	U	B	O	I	S
R6	M	E	N	T	O	A	I	D	X	X

: TROOPS HEADING WEST NEED MORE SUPPLIES SEND
GENERAL DUBOIS MENT TO AID

ii) keyword = SECRET

C1 = ILEDT NHPTK SOJIT MAXSA XERXT ANONI SNEOT X

$$R = \frac{36}{6} = 6$$

R1	I	H	I	S	T	S	R1	T	H	I	S	S	
R2	L	P	I	A	A	N	R2	A	P	L	A	I	N
R3	E	T	T	X	N	F	R3	N	T	E	X	T	F
R4	D	R	M	E	O	O	R4	O	R	D	E	M	O
R5	T	S	A	R	N	T	R5	N	S	T	R	A	T
R6	N	O	X	X	I	X	R6	I	O	N	X	X	X

: THIS IS A PLAIN TEXT FOR DEMONSTRATION

iii) Keyword - PLAIN

CI = VEY EAX ARA ATK HGD

$$R = \frac{15 \times 3}{5}$$

⑤ ③ ① ② ④
P L A I N

R1 H A V E A
R2 G R E A T
R3 D A Y X X

PT: HAVE A GREAT DAY

Permutation:

Find the inverse permutation for

1) 4 3 1 5 2

: $\frac{1 \ 2 \ 3 \ 4 \ 5}{4 \ 3 \ 1 \ 5 \ 2} \rightarrow \text{INDEX}$

Inverse: 3 5 2 1 4

2) 7 4 1 3 5 2 6

: $\frac{1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7}{7 \ 4 \ 1 \ 3 \ 5 \ 2 \ 6} \rightarrow \text{INDEX}$

Inverse: 3 6 4 2 5 7 1

3) 4 1 3 2

: $\frac{1 \ 2 \ 3 \ 4}{4 \ 1 \ 3 \ 2} \rightarrow \text{INDEX}$

Inverse: 2 4 3 1

4) 3 5 1 4 2

: $\frac{1 \ 2 \ 3 \ 4 \ 5}{3 \ 5 \ 1 \ 4 \ 2} \rightarrow \text{INDEX}$

Inverse: 3 5 1 4 2